Experimental Study of the Mach-Zehnder and Michelson Interferometers

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Fundamental quantum principles, such as interference, form the basis for studying in quantum information science(QIS). This article presents the experimental setup, results, and analysis of two classical optical interferometers: the Mach-Zehnder Interferometer (MZI) and the Michelson Interferometer. Here we describe the interferometer setup, experimental procedure and analyze experimental results.

I. INTRODUCTION

Quantum information science (QIS) leverages principles of quantum mechanics and manilulate information, paving the way for advancements in computing, communication, and measurement technologies. Centeral to QIS are Fundamental quantum principles, e.g. interference. Interferometers utilize the wave-like nature of light to investigate phenomena such as phase shifts and interference.

Among the most widely studied interferometers are the Mach-Zehnder interferometer (MZI) and the Michelson Interferometer, both of which offer unique advantages in experimental setups and applications. The MZI is known for its ability to distinguish between different paths taken by light, making it an essential tool for exploring coherent superpositions and measureing phase shifts with high precision. In contrast, the Michelson Interferometer provides valuable insights into changes in optical path lengths, allowing for the detection of minute variations in length and providing a framework for numerous applications in metrology and sensing.

This article presents a preliminary investigation of there two classical optical interferometers.

In part two, we investigate the experimental procedures of the MZI in detail. Quantitative measurements of MZI's two outputs are discussed. Path information introduced by polirization is examined via experiments the result of which is analyzed.

In part three, the setup and experimental procedure of Michelson Interferometer is investigated. Experiments that quantifies misalignment are conducted, the result of which is displayed and analyzed in this part.

Through this investigation, we aim to elucidate the principles and practical applications of both the MZI and Michelson Interferometer, providing valuable insights into their operation and significance in quantum optic lab.

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II. MACH-ZEHNDER INTERFEROMETER

The Mach-Zehnder Interferometer (MZI) is a pivotal instrument in optical physics and engineering, renowned for its ability to measure phase shifts in coherent light beams with high precision. Developed by Ludwig Mach and Hans Zehnder in the early 20th century, the MZI operates on the principle of interference, wherein two light paths are compared to detect minute changes in the optical path length due to external influences, such as temperature variations, pressure changes, or refractive index alterations[1].

The MZI consists of two beam splitters and two mirrors arranged to create two distinct optical paths. When a coherent light source, such as a laser, is introduced, it is split into two beams by the first beam splitter. These beams travel along separate paths and are subsequently recombined at the second beam splitter. The interference pattern produced at the output is highly sensitive to any phase changes introduced along the paths, allowing for the quantitative measurement of various physical phenomena[2].

Applications of the MZI are extensive, ranging from fundamental research in quantum mechanics and optics to practical uses in telecommunications, sensors, and biomedical imaging. Its ability to provide precise measurements makes the MZI an essential tool in experimental physics, particularly in studies involving wavefront manipulation and phase-sensitive detection[3].

A. Establishing the Operational Configuration of the MZI

1. *The optical architecture of MZI*

To construct a Mach-Zehnder Interferometer (MZI), we first establish a stable optical bench to minimize vibrations and external disturbances. The setup begins with a coherent light source, i.e.laser, positioned at one end of the bench. The beam is directed toward a beam splitter via two steering mirror, where it is divided into two distinct paths.

Steering mirrors are strategically placed to guide the beams along their respective paths. These mirrors are mounted on adjustable stages to allow for precise alignment. The light paths are carefully adjusted to ensure that the beams will eventually converge at a second beam splitter.

After interacting with any necessary optical elements, the beams are directed towards the second beam splitter, where they are recombined. The interference patterns generated provide insights into the phase relationships between the beams.

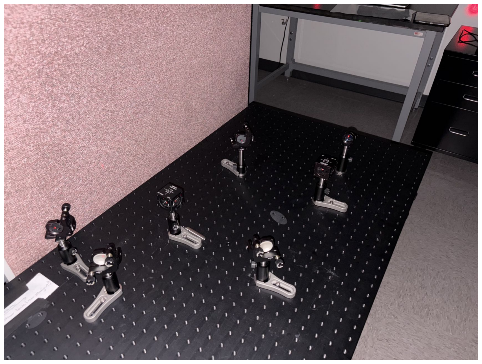


FIG. 1. the operational MZI configuration

To optimize the alignment, iterative adjustments of the steering mirrors are performed, ensuring that both beams overlap at the detector. This meticulous setup is crucial for achieving high visibility and contrast in the interference fringes, allowing for accurate measurements and analysis of the optical characteristics of the system.

2. Interference pattern

Interference patterns is a fundamental phenomenon that occurs when two or more coherent waves overlap, resulting in regions of constructive and destructive interference. Constructive interference occurs when wave crests or troughs align, producing bright or amplified regions. Destructive interference happens when a crest meets a trough, canceling each other out and forming dark or diminished regions.

The interference of light is distinctive in that the superposition of electromagnetic (EM) fields cannot be directly observed, unlike the superposition of waves in water. Instead, the superposition of EM fields is a theoretical construct essential for explaining how two light beams can intersect and continue propagating along their original trajectories without disruption. Interferometer is a prominent example of light interference.

As opposed to mechanical waves, optical waves are complex valued wave functions. Complex valued wave functions are needed to illustrate the interference of light waves.

Assume the displacement of two light waves at point z is:

$$U_1(z,t) = A_1(\mathbf{z})e^{i[\varphi_1(z)-\omega t]} \tag{1}$$

$$U_2(z,t) = A_2(\mathbf{z})e^{i[\varphi_2(z)-\omega t]} \tag{2}$$

Here A represents the amplitude of displacement, ϕ represents the phase and the ω represents the angular frequence.

The summed displacement is

$$\begin{aligned} U(z,t) &= A_1(z)e^{i[\varphi_1(z)-\omega t]} + A_2(z)e^{i[\varphi_2(z)-\omega t]} \\ &= [A_1(z) + A_2(z)e^{i[\varphi_1(z)-\varphi_2(z)]}]e^{i[\varphi_1(z)-\omega t]} \end{aligned} \tag{3}$$

From the functional relationship, it is evident that the interference pattern reflects the phase difference between the two waves, with maxima occurring when the phase difference is an integer multiple of 2π. For two beams of equal intensity, the maxima exhibit an intensity four times that of the individual beams, while the minima correspond to zero intensity.

From this framework, we can derive the mathematical model for the interference pattern produced by two incident beams that are slightly misaligned. This configuration simplifies the experimental setup, making the pattern easier to generate and observe.

The setup is misaligned intentionally to create vertical fringes. Figure 2 and 3 display the interference patterns acquired from MZI.

FIG. 2. Interference pattern generated by the experimental setup, demonstrating the effects of interference

B. Quantitative measurement

In a properly aligned Mach-Zehnder Interferometer (MZI), a significant difference in the output intensities is observed between the two output ports. This difference arises from the interference of the two coherent beams, where constructive interference occurs at one port, resulting in higher intensity, and destructive interference occurs at the other port, leading to minimal or zero intensity.

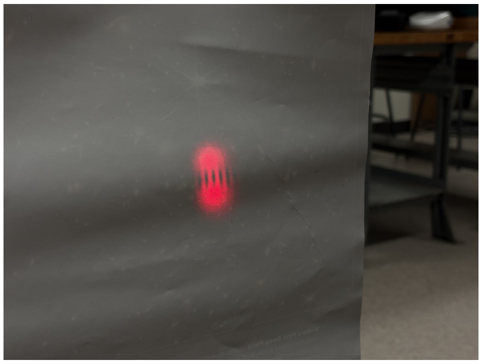


FIG. 3. Interference pattern emerging from the second output port

It is worth noting that upon reflection from a beam splitter, the light beam undergoes a relative phase shift of π radians. This phenomenon arises due to the discontinuity introduced at the boundary of the beam splitter, which can be likened to the effect observed when light reflects from a medium of higher refractive.[4]

Following meticulous alignment of our MZI, we conducted a numerical measurement of the optical power output from both ports. This experimental procedure yielded the voltage readings shown in TABLE.I

By incorporating measured data into the function provided below, we can calculate the contrast of the outputs. The contrast C is defined as:

$$C = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} \tag{4}$$

Our MZI configuration yielded 56.5% contrast between output beams.

C. Wavefront geometries

1. *Experimental results*

The geometry of wavefronts—specifically whether they are planar or spherical—significantly influences the interference patterns produced in a MZI. When plane waves are incident on the interferometer, they yield uniform interference patterns characterized by evenly spaced bright and dark fringes. This uniformity arises from the consistent phase differences across the wavefront, leading to predictable constructive and destructive interference outcomes. Conversely, when spherical wavefronts are present, the interference pattern can become more intricate due to the variation in path lengths and phase shifts across the wavefront. The resulting irregular fringe spacing and intensity distributions complicate the interference pattern, making it less uniform than that produced by plane waves.

TABLE I. Voltage measurements were obtained for the two output beams of the MZI, designated as Output 0 for the beam parallel to the input beam and Output 1 for the beam perpendicular to the input beam.

Output 0	Output 1
9.00V	2.55V

We conduct experiments to investigate the interference phenomena involving wavefront configurations: the interaction of two plane waves, the interference of one plane wave with one spherical wave, and the interference between two spherical waves.

To generate spherical wavefronts, we position a convex lens downstream of the mirrors and upstream of the downstream beamsplitter in the optical configuration.

In Fig2 and Fig3, the interference pattern of two plane wavefront is displayed.

We generate the interference pattern resulting from the interaction of one spherical wavefront and one plane wavefront by introducing a convex lens with a focal length of 100 mm into one of the arms of the MZI. The interference pattern of this modified setup is displayed in fig4

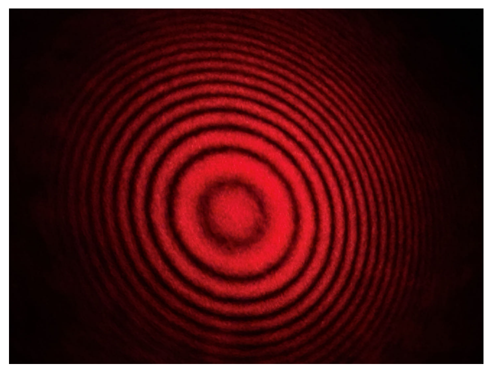


FIG. 4. the interference pattern of plane wavefront interacting with spherical wavefront

2. *Numerical calculation*

By neglecting factors such as amplitude and polarization, the interference phenomenon can be mathematically modeled using the following framework.[5]

$$\begin{aligned} E &\sim \cos(kz - \omega t) \\ &\quad + \cos(kz + k(x^2 + y^2)/[2R(z)] - \omega t) \end{aligned} \tag{5}$$

where $R(z)$ is the radius of curvature of the spherical wavefront at z position. It can be calculated by measureing the radii of fringes:

$$r_m = (2\lambda R m)^{1/2} \tag{6}$$

In our experiment, the radius of the second-order fringe ($m = 2$), r_2, was measured to be 17 mm. This implies that the radius of curvature at this point, given the experimental setup, is 114.13 m.

We generate the interference pattern resulting from the interaction of two spherical wavefront by introducing one convex lens with a focal length of 100 mm and one convex lens with a focal length of 150mm into the arms of the MZI. The interference pattern under this configuration is given in Fig.5

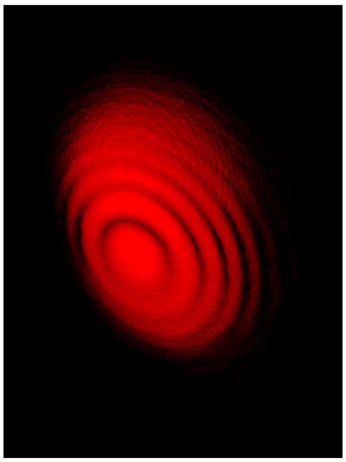


FIG. 5. the interference pattern of interaction between two spherical wavefront with different radius of curvature

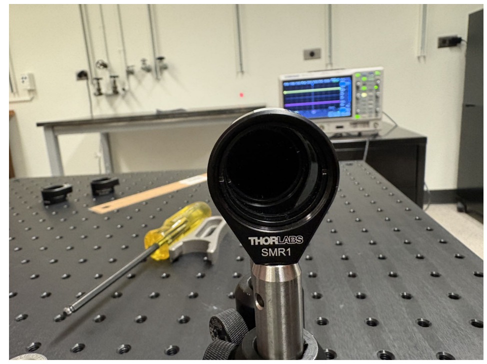


FIG. 6. a demonstration of two orthogonal polarization lenses

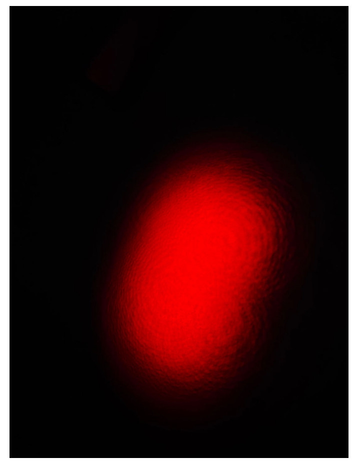


FIG. 7. Pattern of lasers combined from two polarization labeled paths

D. Path information

1. *Experimental results*

A MZI generates interference patterns when a laser, is divided into two paths and subsequently recombined. However, introducing the ability to determine the path taken by individual photons, known as a "which-way" measurement, results in the disappearance of interference fringes. This phenomenon is a fundamental principle in quantum mechanics.

In this experiment we introduce path information by polarization. By adding two polarization lens that are orthogonal, demonstrated in Fig.6, between mirrors and downstream beamsplitter. The interference patterns obtained by this optical configuration is displayed in fig7. It is clear that no interference is obtained by combining two labeled lasers.

Now, by introducing an additional rotating polarization lens, the analyzer, subsequent to the downstream beam splitter, we can observe the interference pattern for polarization angles of 0, $\pi/4$, $\pi/2$. These results are summarized in tableII

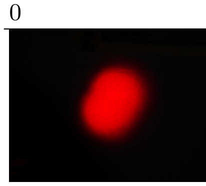
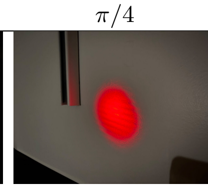
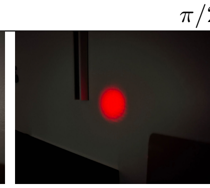
No discernible interference patterns were observed when the analyzer was oriented at 0 or $\pi/2$. However, interference patterns became evident when the polarizer was rotated to $\pi/4$.

2. *Theoretical analysis*

A classical field analysis and a quantum analysis and its interpertation is given in M. Schneider and I. LaPuma's article[6].

Classical wise, the sum of two polarized laser can be

TABLE II. Rotate

0	$\pi/4$	$\pi/2$
		

written as:

$$\begin{aligned}\mathbf{E}_{net} &\approx E_0 e^{-i\omega t}[\hat{x}e^{i(k_x x+k_z z)} + \hat{y}e^{i(-k_x x+k_z z)}] \\ &= E_0 e^{i(k_z z-\omega t)}[\hat{x}e^{ik_x x} + \hat{y}e^{-ik_x x}]\end{aligned} \tag{7}$$

The Intensity of this wave can by simplified to:

$$E_x^2 + E_y^2 \propto E_0^2|e^{ik_x x}|^2 + E_0^2|e^{-ik_x x}|^2 = 2E_0^2 \tag{8}$$

that is, a constaint.

Following the analyzer rotated to a $\pi/4$ angle, the resulting electric field is

$$\begin{aligned}\mathbf{E}_{analyzed} &\approx E_0 e^{-i\omega t}[\hat{d}e^{i(k_x x+k_z z)} + \hat{d}e^{i(-k_x x+k_z z)}] \\ &= 2E_0\hat{d}e^{i(k_z z-\omega t)}\cos(k_x x),\end{aligned} \tag{9}$$

The intensity now displays a repeating sequence of bright and dark bands, characteristic of an interference pattern.

A quantum interpretation is presented.

Assume the previously mentioned orthogonal polarizations are denoted as $|x\rangle$ and $|y\rangle$, incident beam can be discribed as

$$|\Psi_i\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}} \tag{10}$$

ouput beam of the polarizing interferometer Configuration is then

$$|\Psi_o\rangle = \frac{|x\rangle + e^{i\phi}|y\rangle}{2} \tag{11}$$

The analyzer at $\pi/4$ affects on the polarization basis analogous to a Hadamard gate

$$\hat{P}_{\pi/4}|x\rangle = \frac{|x\rangle + |y\rangle}{2}, \quad \hat{P}_{\pi/4}|y\rangle = \frac{|x\rangle + |y\rangle}{2} \tag{12}$$

The output beam of the analyzer can be derived

$$\begin{aligned}&\langle\Psi_{after}|\hat{P}_{\pi/4}|\Psi_{after}\rangle \\ &= \langle\Psi_{after}|\frac{(|x\rangle + |y\rangle) + e^{i\phi}(|x\rangle + |y\rangle)}{4} \\ &= \frac{1+\cos(\phi)}{2},\end{aligned} \tag{13}$$

Which is in the form of a interference fringe.

E. Modification on the MZI Configuration

By substituting the downstream beamsplitter, we created a modified interferometer. Through careful analysis of the optical path, we determined that the output of the remaining beamsplitter's port, orthogonal to the input beam, generates an interference pattern akin to that observed in the original MZI. Experimental findings support this conclusion.

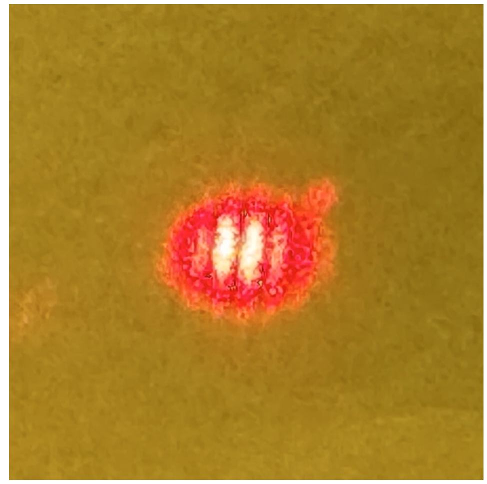


FIG. 8. output of the modified MZI

III. THE MICHELSON INTERFEROMETER

The Michelson interferometer, a cornerstone instrument in optics, is a precision device used to measure distances with extraordinary accuracy. Invented by Albert A. Michelson in 1881[7], it has played a crucial role in numerous scientific discoveries, including the famous Michelson-Morley experiment that challenged the concept of a luminiferous ether.

The interferometer splits a beam of light into two paths using a partially silvered mirror, known as a beamsplitter. Each beam travels a different distance before being reflected back to the beamsplitter, where they recombine and interfere. The resulting interference pattern, observed as a series of bright and dark fringes, is highly sensitive to changes in the path length difference between the two beams.

By carefully controlling the path length difference, scientists can measure minute displacements, refractive indices, and other optical properties with remarkable precision. The Michelson interferometer remains an indispensable tool in fields such as metrology, spectroscopy, and optical coherence tomography.[8]

A. Establishing the operational configuration of Michelson Interferometer

To construct a Michelson interferometer, a stable optical table is essential to minimize vibrations that can disrupt the interference pattern. A monochromatic light source, i.e.laser, is employed to provide coherent illumination. A beamsplitter divides the incident beam into two paths of equal intensity.

Each beam is then directed towards a mirror, one of which is mounted on a precision translation stage to allow for precise path length adjustments. Upon reflection, the beams recombine at the beamsplitter, interfering constructively or destructively. This interference pattern is observed on a screen or detected by a suitable detector.

Careful alignment of the optical components is crucial. The mirrors must be perpendicular to the incident beams, and the beam paths should be as parallel as possible. By adjusting the position of the movable mirror, the path length difference between the two arms can be varied, leading to a shift in the interference pattern.

The resulting interference pattern, a series of bright and dark fringes, can be analyzed to measure distances with high precision, study the properties of light, and investigate various optical phenomena.

We construct an operational configuration of Michelson interferometer as Fig9 One of the reflecting mirrors

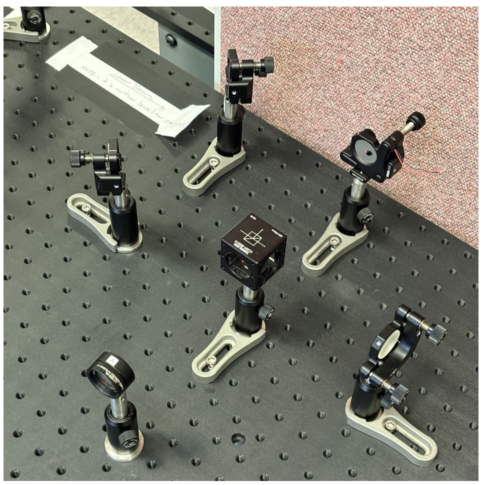


FIG. 9. an operational configuration of Michelson Interferometer

is attached to a piezo-electric mount, facilitating precise positional control through an input electrical signal.

From this particular configuration we obtained interference patter depicted in Fig10

FIG. 10. interference pattern from Michelson Interferometer

B. Quantifying misalignment and estimating coherence length

In this part we will utilize the current Michelson interferometer to introduce known misalignments and measure the resulting fringe contrast as a function of path length difference. By analyzing these measurements, we will determine the coherence length of the laser.

We conduct experiment with three different beamsplitter configuration, the misalignment are respectively 0.17, 3.4 and 6.8 mrad. Result data can be found in GitHub Fig11 illustrates the contrast at different arm length and misalignment.

From a theoretical standpoint, it is expected that the gradient of the fitted function will be greater for larger misalignments. Our empirical findings are generally consistent with this hypothesis.

Extrapolation of the data presented in Fig12 to the limit of zero misalignment suggests that the laser's intrinsic coherence length is on the order of 149.44 mm.

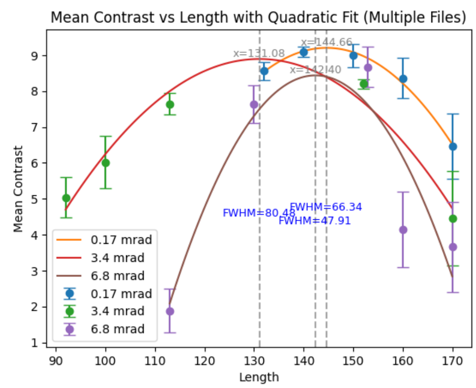


FIG. 11. an operational configuration of Michelson Interferometer

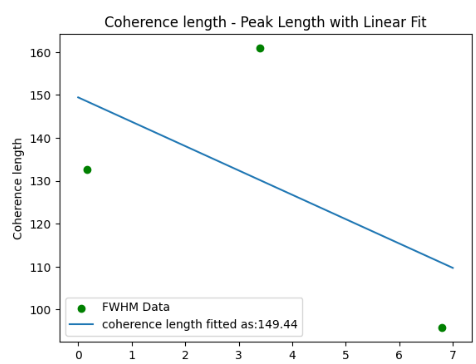


FIG. 12. coherence length at each misalignment fitted to find coherence length at 0 rad misalignment

IV. CONCLUSION

This lab session delved into the fundamental principles of interferometry, specifically focusing on the Mach-Zehnder Interferometer (MZI) and Michelson Interferometer. The MZI was meticulously constructed and aligned, enabling the observation and quantification of interference patterns. By varying the optical configuration of the MZI, distinct interference patterns were generated and analyzed, validating the theoretical predictions derived from the mathematical model.

Furthermore, the experiment explored the concept of quantum erasure, demonstrating how the visibility of interference fringes can be influenced by the acquisition of path information. The Michelson Interferometer was employed to measure the coherence length of the laser source. By analyzing the fringe contrast as a function of the optical configuration, an estimate for the coherence length was obtained, along with an associated error analysis.

The outcomes of this experiment provide valuable insights into the behavior of light waves and the underlying principles of quantum mechanics. The experimental techniques and data analysis methods presented here can be applied to a wide range of optical and quantum optics experiments.

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